

This unit is assessed by means of a written examination paper of 1 hour 20 minutes' duration. The paper will consist of short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not seen before.

The total number of marks available for this examination paper is 40. It contributes 10 per cent to the International Advanced Level in Physics.

Students will need a scientific calculator, a protractor and a ruler for this paper.

Students are provided with the formulae sheet in *Appendix 5* in the main specification. Any other physics formulae that are required will be stated in the question paper.

10.3 Assessment details

This unit is assessed by means of a written examination paper. The assessment for this unit covers planning an experiment, analysing data and drawing conclusions. A laboratory is not required for this assessment.

Planning

Students will be expected to plan an experiment that is set by Pearson, although they will not be expected to carry it out.

Students may be required to:

- identify the most appropriate apparatus, giving details. These may include the range and precision of instruments and/or relevant dimensions of apparatus (e.g. the length of string used for a pendulum)
- discuss calibration of instruments, e.g. whether a meter reads zero before measurements are made
- describe how to measure relevant variables using the most appropriate instrument(s) and techniques
- identify and state how to control all other relevant variables to make it a fair test
- discuss whether repeat readings are appropriate
- identify health and safety issues and discuss how these may be dealt with
- discuss how the data collected will be used
- identify possible sources of uncertainty and/or systematic error and explain how these may be reduced or eliminated.

Implementation and measurements

Students will be given partial details of how an experiment was carried out. Results may be included.

Students may be asked to:

- comment on how the experiment could have been improved, possibly by using additional apparatus (e.g. to reduce errors) — examples may include using set squares to measure the diameter of a cylinder and using a marker for timing oscillations
- comment on the number of readings taken
- comment on the range of measurements taken
- comment on significant figures — students may be required to identify and/or round up any incorrect figures in a table of results
- identify and/or amend units that are incorrect

Analysis

Students will be expected to use the set of experimental results to:

- perform calculations, using the correct number of significant figures
- plot results on a graph using an appropriate scale and units — the graph could be logarithmic in nature
- use the correct units throughout
- comment on the trend/pattern obtained
- determine the relationship between two variables or determine a constant with the aid of the graph, e.g. by determining the gradient using a large triangle
- use the terms *precision*, *accuracy* and *sensitivity* appropriately
- suggest realistic modifications to reduce errors
- suggest realistic modifications to improve the experiment
- discuss uncertainties qualitatively and quantitatively
- compound percentage uncertainties correctly.

Drawing conclusions

After processing results, students may be asked to provide a final conclusion for the experiment in relation to its original aim and based on their quantitative evidence.

They may also be asked to suggest further relevant work, for example to verify their conclusion(s).